Drawing to Documentation by Gemini [Drawing ta bojhar jonno]

ClaritySync: RDBMS Final Planning Documentation

Legend

- **Pink:** Page / UI level changes
- **Blue:** Database (DB) level changes
- **Yellow:** System level changes/logic

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1. Project Status: Done

- **Contacts Module:**
  - Categorized into Buyers, Sellers, and Both.
  - Contact details tracking.
  - Transaction history.
  - Due adjustments (Note: Needs review/adjustment, potentially no DB changes required).

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2. To-Do: Employee Modules

- **Add Employee:** Functionality to add new employees.
- **List Employees:** Display an employee directory.
- **System Access:** Give system access by creating user accounts linked to employees.

- **Payroll:** Payslip and salary-related features.
- **Attendance Tracking:** Record Present, Absent, and Leave statuses.
- **Transaction History:** Log employee-related transactions.
- **Session Tracking:** Database ID / session track (requires DB and state management).
- **Reports:** Attendance and payslip reports.

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3. Enhancements for "Done" Modules

- **System:** Products can now have a manufacture date.
- **DB:** Ensure the field is marked as `Not Null`.
- **Page/UI:** Add this new manufacture date field in both the "Product Setup" and "Add Stock" pages.

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4. Module Setup & System Settings

Toggleable Features to Implement:

- Warranty service
- Employee attendance
- User access/manager module

Settings (System) Requirements:

- Full system user roles (tree graph/hierarchy).
- System log / details (with email).
- Category manager *(Marked as Done)*.
- Tax setup.
- System rule configurations.
- Flag bar for toggling features.
- **Activity Log:** Log all actions of the user. Track username, IP address, action description, and session validity.

5. Setup Process / Flow

- *Question:* Do we need to do this?
- **DB:** System initialization process (`sys init`) to create a default admin user upon first launch.

6. Warranty Module (Tab)

- **DB Policy Enforcements:** We need to register/enforce warranty policies manually.
- **Damaged Items:** Serial numbers of damaged items must be stored in the DB and marked as "damaged/discarded".
- **Exchange Flow:**
  - Only replace damaged items with new ones if they are within a valid warranty period.
  - *Strategy Decision:* If exchanged, will the current warranty continue, or will the new item get a newly issued warranty time limit? (Requires deciding on an applicable strategy from the referenced documentation).
  - Need to finalize how the return flow will be maintained overall.
- **Warranty Period Start Logic:**
  - When does it start? Sale date? Purchase date? Manufacture date?
  - *Decision:* Give the user the ability to change the warranty start logic on a per-product basis.

7. Architecture / Workflow

1. **Add Product:** Setup when the warranty countdown will start. This is especially important for products that spoil or deteriorate over time, like batteries.
2. **Sells Product:** The system should trigger an alert if a product's warranty is already expiring (e.g., if the countdown started before the sale).

3. **Product Damaged & Exchanged:** A new item is issued. This new item may have its own valid warranty period.
4. **Discard/Return:** The damaged item is either discarded or exchanged with the manufacturer for free replacement items.